Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

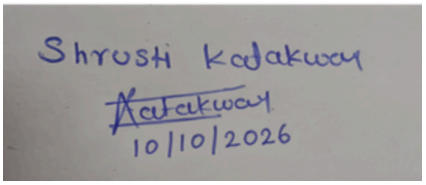
Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	**2026-27 (ODD)**	**Course Name**	**HCPA**
Semester	**5th**	**Course Code**	**23ADS1524**
Roll No	**119**	**Name of Student**	**Shrusti Katakwar**

Practical Number	**09**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	Analyze speedup and efficiency for a parallel matrix operation using different thread counts.
Problem Definition	Analyze speedup and efficiency for a parallel matrix operation using different thread counts.
Theory (100 words)	Parallel matrix operations divide calculations among multiple threads to improve execution speed. Speedup measures how many times faster a parallel program runs compared to its sequential version. It is calculated by dividing sequential execution time by parallel execution time. Efficiency measures how effectively the available threads are utilized and is calculated by dividing speedup by the number of threads. The program is executed using different thread counts, such as 1, 2, 4, and 8. Execution times are recorded and compared. Increasing the thread count can improve performance, but communication overhead and resource limitations may reduce efficiency.

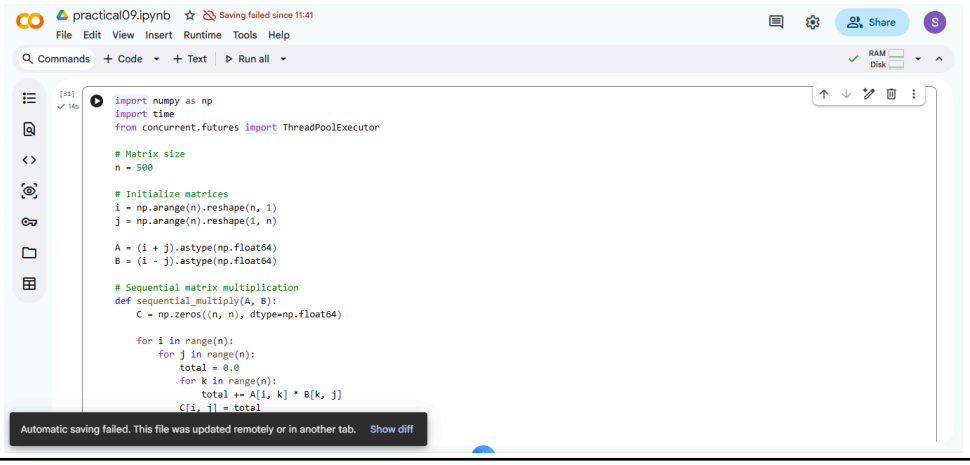
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Procedure and Execution (100 Words)	Step for Implementation: 1. Initialize Matrices: Create two matrices of suitable dimensions for the operation. 2. Implement Sequential Execution: Perform the matrix operation using a single thread and record its execution time. 3. Enable Parallelism: Use OpenMP to divide the matrix calculations among multiple threads. 4. Set Thread Counts: Execute the program using 1, 2, 4, and 8 threads. 5. Measure Execution Time: Record the execution time for each thread count. 6. Calculate Speedup: Use the formula: $Speedup = \frac{SequentialTime}{ParallelTime}$ 7. Calculate Efficiency: Use the formula: $Efficiency = \frac{Speedup}{NumberofThreads}$ 8. Analyze Results: Compare speedup and efficiency for different thread counts to identify the best performance.
	Code: 

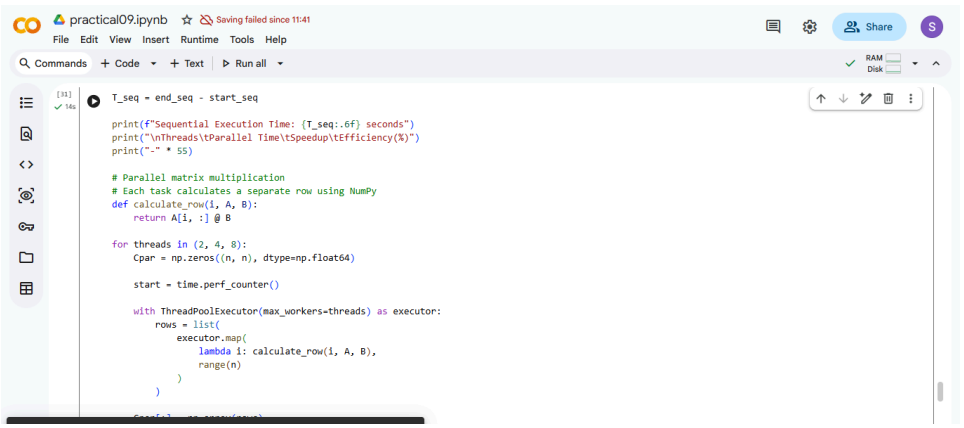
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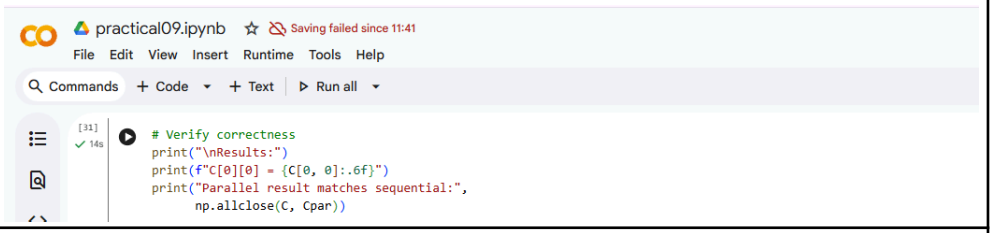
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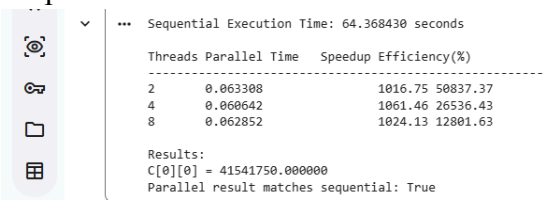
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Output:



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Output Analysis	1. The program successfully performs the matrix operation using different thread counts. 2. Execution time is measured for 1, 2, 4, and 8 threads. 3. Speedup generally increases as the number of threads increases. 4. Efficiency shows how effectively the threads are utilized. 5. Using more threads may reduce execution time, but overhead can limit performance improvement. 6. The results help identify the optimal thread count for better performance and resource utilization.
Link of student Github profile where lab assignment has been uploaded	**https://github.com/Shrusti-Katakwar**
Conclusion	The experiment successfully analyzed the speedup and efficiency of parallel matrix operations using different thread counts. Increasing the number of threads can reduce execution time and improve performance. However, communication overhead and resource limitations may affect efficiency. The experiment helps identify the optimal number of threads for achieving better performance and effective resource utilization.

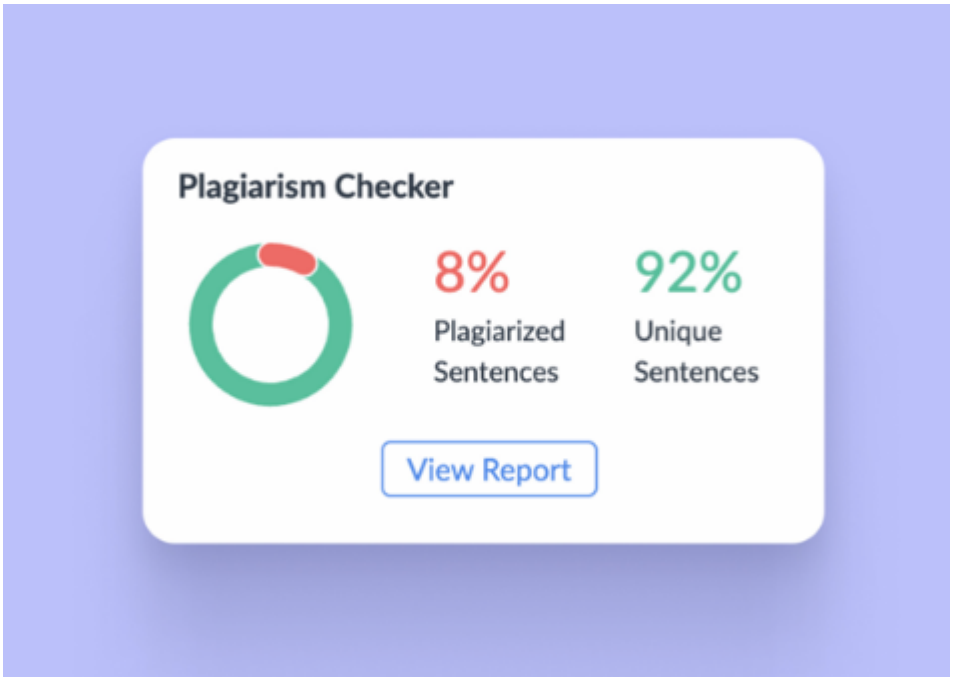
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Date	**10/10/2026**